

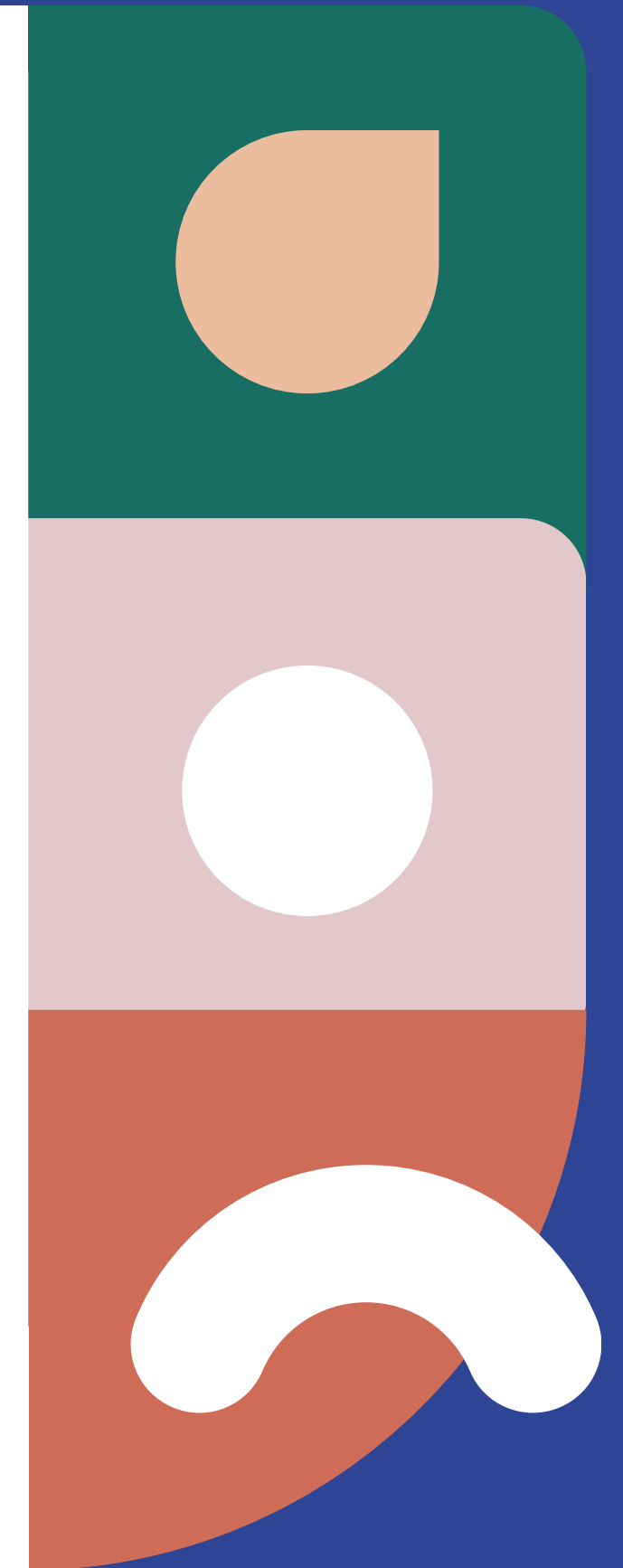


MySawit

By Advprog A18

Member Group

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Link Repository

Frontend

<https://github.com/advprog-2026-A18-project/frontend-mySawit>

API Gateway

<https://github.com/advprog-2026-A18-project/API-Sawit-Gateway>

Autentikasi

<https://github.com/advprog-2026-A18-project/Autentikasi-Autorisasi-Manajemen-Pengguna>

Manajemen Kebun Sawit

https://github.com/advprog-2026-A18-project/MySawit-Manajemen_Kebun_Sawit

Manajemen Hasil Panen Sawit

<https://github.com/advprog-2026-A18-project/Manajemen-Hasil-Panen-Sawit>

Manajemen Pengiriman Panen Sawit

<https://github.com/advprog-2026-A18-project/Manajemen-Pengiriman-Hasil-Panen-Sawit>

Manajemen Pembayaran

<https://github.com/advprog-2026-A18-project/Manajemen-Pembayaran>

Deployment

<https://github.com/advprog-2026-A18-project/MySawit-Deployment>

Monitoring

<https://github.com/advprog-2026-A18-project/Monitoring-server>

Link Deployment

Frontend

<https://mysawit-frontend-2d6fbc97d3fe.herokuapp.com/>

Autentikasi

<http://3.211.173.168:8080/auth-service>

Manajemen Kebun Sawit

<http://3.211.173.168:8080/manajemen-kebun-sawit-service/>

Manajemen Hasil Panen Sawit

<http://3.211.173.168:8080/manajemen-hasil-panen-sawit-service/>

Manajemen Pengiriman Panen Sawit

<http://3.211.173.168:8080/pengiriman-hasil-panen-sawit-service/>

Manajemen Pembayaran

<http://3.211.173.168:8080/pembayaran-service/>

Link SonarQube

SONAR

<https://sonarcloud.io/organizations/advprog-2026-a18-project/projects>



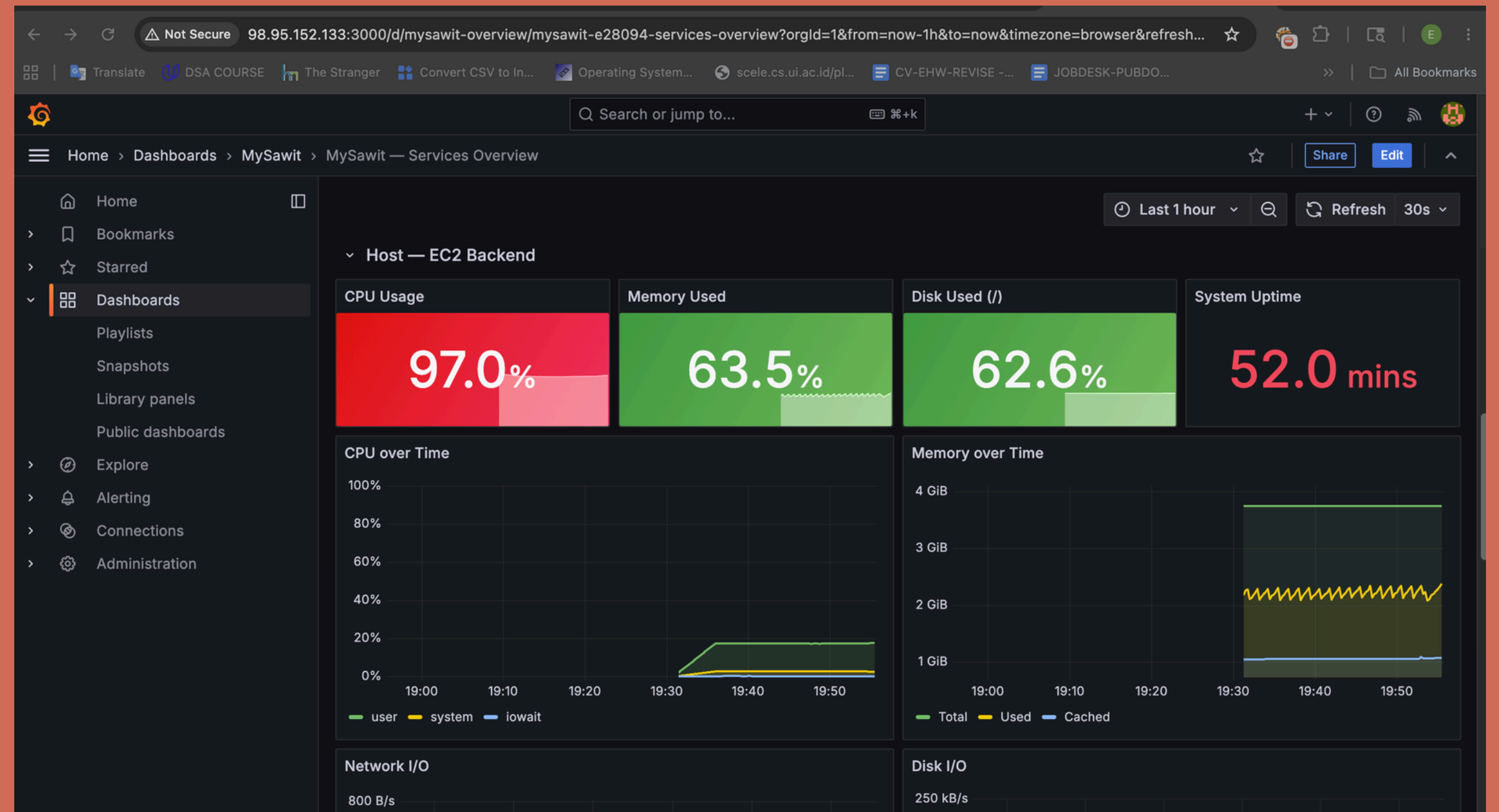
Link Monitoring

MONITORING

<http://98.95.152.133:3000/>

Credential:

- Username: admin
- Password: Halo__#12



Software Design

Design Pattern (BUILDER)

```
@Data @evanhwz
@NoArgsConstructor
@AllArgsConstructor
@Builder
public class UserResponseDTO {
    private UUID id;
    private String username;
    private String fullname;
    private String email;
    private String role;
    private String nomorSertifikasi; // null if user not mandor
    private String namaMandor;
}
```

```
public UserResponseDTO toDto(User user, String nomorSertifikasi) { 3 usages @evanhwz
    String namaMandor = user.getMandor() != null
        ? user.getMandor().getFullname()
        : null;

    return UserResponseDTO.builder()
        .id(user.getId())
        .username(user.getUsername())
        .fullname(user.getFullname())
        .email(user.getEmail())
        .role(user.getRole() != null ? user.getRole().getName() : null)
        .nomorSertifikasi(nomorSertifikasi)
        .namaMandor(namaMandor)
        .build();
}
```

Software Design

Design Pattern (Static FACTORY)

```
public class ApiResponse<T> {

    public static <T> ApiResponse<T> badRequest(String message, T data) { 1 usage  & evanhwz
        return of(ApiResponseStatus.BAD_REQUEST, message, data);
    }

    public static <T> ApiResponse<T> unauthorized(String message) { 4 usages  & evanhwz
        return of(ApiResponseStatus.UNAUTHORIZED, message, data: null);
    }

    public static <T> ApiResponse<T> forbidden(String message) { 1 usage  & evanhwz
        return of(ApiResponseStatus.FORBIDDEN, message, data: null);
    }

    public static <T> ApiResponse<T> notFound(String message) { return of(ApiResponseStatus.NOT_FOUND, message, data: null);

    public static <T> ApiResponse<T> internalServerError(String message) { 2 usages  & evanhwz
        return of(ApiResponseStatus.INTERNAL_SERVER_ERROR, message, data: null);
    }

    public static <T> ApiResponse<T> of(HttpStatus status, String message, T data) { & evanhwz
        return new ApiResponse<>{
            status.value(),
            message,
            data
        };
    }

    public static <T> ApiResponse<T> of(ApiResponseStatus status, String message, T data) { & evanhwz
        return new ApiResponse<>{
            status.getCode(),
            message != null ? message : status.getDefaultMessage(),
            data
        };
    }
}
```

```
String refreshToken = refreshTokenService.createRefreshToken(registeredUser.user().getId()).getToken();
AuthResponseDTO authResponseDTO = authResponseMapper.toDto(
    registeredUser.user(),
    registeredUser.nomorSertifikasi(),
    refreshToken
);
return ApiResponse.created(message: "Google registration succeed! You are authenticated", authResponseDTO);
```

Software Design

S.O.L.I.D (Single Responsibility Principle)

```
@RestController @ evanhwz
@RequestMapping("/auth")
@RequiredArgsConstructor
public class AuthController {

    private final LocalAuthService localAuthService;
    private final OAuthService<GoogleAuthRequestDTO> googleAuthService;

    private final RefreshTokenService refreshTokenService;

    @PostMapping("/register") @ evanhwz
    public ResponseEntity<ApiResponse<AuthResponseDTO>> register(@Valid @RequestBody RegisterRequestDTO request){
        ApiResponse<AuthResponseDTO> response = localAuthService.register(request);
        return ResponseEntity.status(response.getStatusCode()).body(response);
    }

    @PostMapping("/login") @ evanhwz
    public ResponseEntity<ApiResponse<AuthResponseDTO>> login(@Valid @RequestBody LoginRequestDTO request) {
        ApiResponse<AuthResponseDTO> response = localAuthService.login(request);
        return ResponseEntity.status(response.getStatusCode()).body(response);
    }

    @PostMapping("/google") @ evanhwz
    public ResponseEntity<ApiResponse<AuthResponseDTO>> googleLogin(@Valid @RequestBody GoogleAuthRequestDTO request) {
        ApiResponse<AuthResponseDTO> response = googleAuthService.authenticate(request);
        return ResponseEntity.status(response.getStatusCode()).body(response);
    }
}
```

Software Design

S.O.L.I.D
(Open Close Principle)

```
public abstract class ApiException extends RuntimeException { 11 usages 5 inheritors ③ eva  
  
    private final HttpStatus status; 2 usages  
  
    protected ApiException(HttpStatus status, String message) { 5 usages ③ evanhwz  
        super(message);  
        this.status = status;  
    }  
  
    > public HttpStatus getStatus() { return status; }  
}
```

Software Design

S.O.L.I.D (Liskov Substitution Principle)

```
> import ...  
  
public interface OAuthService<T> { 5 usages 1 implementation & evanhwz  
    ApiResponse<AuthResponseDTO> authenticate(T request); 15 usages 1 implementation  
}  
  
public class GoogleAuthServiceImpl implements OAuthService<GoogleAuthRequestDTO> {  
  
    private final UserRepository userRepository;  
    private final RefreshTokenService refreshTokenService;  
    private final AuthResponseMapper authResponseMapper;  
    private final UserRegistrationService userRegistrationService;  
  
    @Value("${GOOGLE_CLIENT_ID}")  
    private String googleClientId;  
  
    @Transactional 15 usages & evanhwz  
    public ApiResponse<AuthResponseDTO> authenticate(GoogleAuthRequestDTO request) {  
        try {  
            GoogleIdToken idToken = verifyGoogleToken(request.getIdToken());  
  
            if (idToken == null) {  
                return ApiResponse.unauthorized("Invalid Google Token");  
            }  
  
            GoogleIdToken.Payload payload = idToken.getPayload();  
            String email = payload.getEmail();  
            String name = (String) payload.get("name");  
            String username = email.split(regex: "@")[0] + "_" + UUID.randomUUID().toString().substring(0, 8);  
  
            Optional<User> userOptional = userRepository.findByEmail(email);  
  
            if (userOptional.isPresent()) {  
                User existingUser = userOptional.get();  
  
                // AUTH PROVIDED GOOGLE equalsTokenPage(existingUser, getAuthProvider())  
  
                return ApiResponse.success(authResponseMapper.map(existingUser, idToken));  
            }  
  
            User newUser = userRegistrationService.register(username, email, name);  
            return ApiResponse.success(authResponseMapper.map(newUser, idToken));  
        } catch (Exception e) {  
            return ApiResponse.unauthorized(e.getMessage());  
        }  
    }  
  
    private final OAuthService<GoogleAuthRequestDTO> googleAuthService;  
  
    @PostMapping("/google") & evanhwz  
    public ResponseEntity<ApiResponse<AuthResponseDTO>> googleLogin(@Valid @RequestBody GoogleAuthRequestDTO request) {  
        ApiResponse<AuthResponseDTO> response = googleAuthService.authenticate(request);  
        return ResponseEntity.status(response.getStatusCode()).body(response);  
    }  
}
```

Software Design

S.O.L.I.D
(Interface Segregation Principle)

```
public interface LocalAuthService { 5 usages 1 implementation 2 evanhwz
    ApiResponse<AuthResponseDTO> register(RegisterRequestDTO request); 10
    ApiResponse<AuthResponseDTO> login(LoginRequestDTO request); 8 usages
}
```

```
public interface OAuthService<T> { 5 usages 1 implementation 2
    ApiResponse<AuthResponseDTO> authenticate(T request);
}
```

© GoogleAuthServiceImpl

① LocalAuthService

© LocalAuthServiceImpl

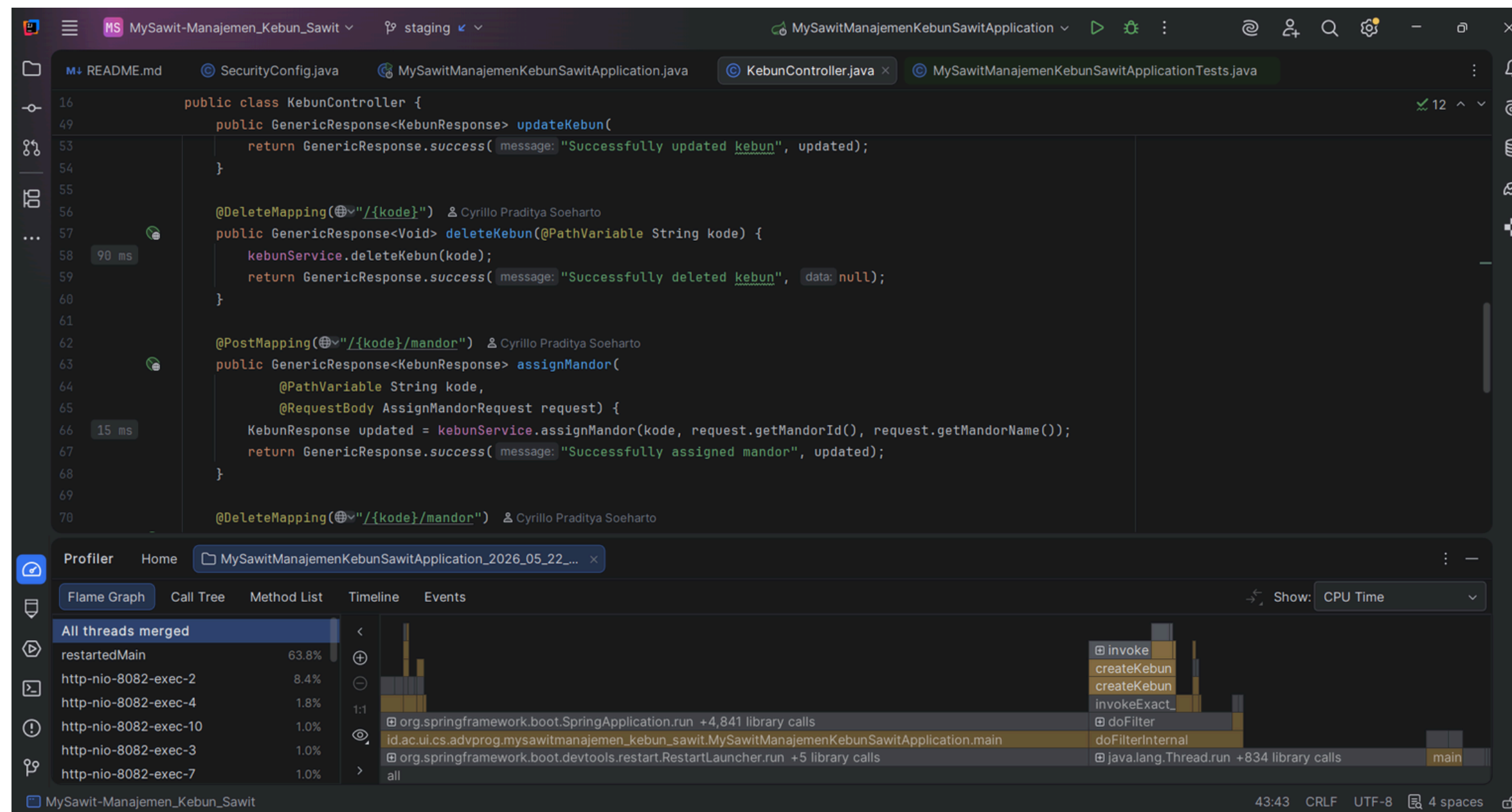
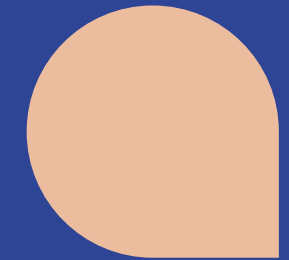
① OAuthService

Software Design

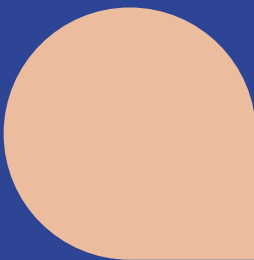
S.O.L.I.D (Dependency Inversion Principle)

```
19 public class PengirimanService {
20
21     private final PengirimanRepository pengirimanRepository;
22     private final EksternalIntegrationService eksternalService;
23
24     //...
27     private static final List<String> STATUS_ORDER = List.of("Memuat", "Mengirim", "Tiba di Tujuan"); 2 usages
28     private static final String STATUS_DISETUUJUI = "DISETUUJUI"; 4 usages
29     private static final Set<String> STATUS_PANEN_DISETUUJUI = Set.of("DISETUUJUI", "APPROVED", "APPROVE"); 1 usage
30     //...
33     @Transactional 16 usages 2 Gebxby
34     public PengirimanResponseDTO tugaskanSupir(CreatePengirimanRequestDTO request, UUID mandorId) {...}
65     //...
68     @Transactional 8 usages 2 Gebxby *
69     public PengirimanResponseDTO updateStatusPengiriman(UUID pengirimanId, UUID supirId, UpdateStatusRequestDTO request) {
93     //...
96     public List<PengirimanResponseDTO> getDaftarPengiriman(String status, UUID supirId, String tanggal) {...}
105
106     public List<PengirimanResponseDTO> getDaftarPengirimanSupir(UUID supirId, String tanggal) { 4 usages 2 Gebxby
107         return pengirimanRepository.findBySupirId(supirId).stream() Stream<Pengiriman>
108             .filter( Pengiriman p -> tanggal == null ||
109                 p.getTanggalPengiriman().toLocalDate().equals(LocalDate.parse(tanggal)))
110             .map(this::convertToResponseDTO) Stream<PengirimanResponseDTO>
111             .collect(Collectors.toList());
112     }
113     //...
116     public List<PengirimanResponseDTO> getDaftarPengirimanMandor(UUID mandorId, String status) {...}
```

Software Quality Profiling



Software Quality Test Coverage



 sawit-panen

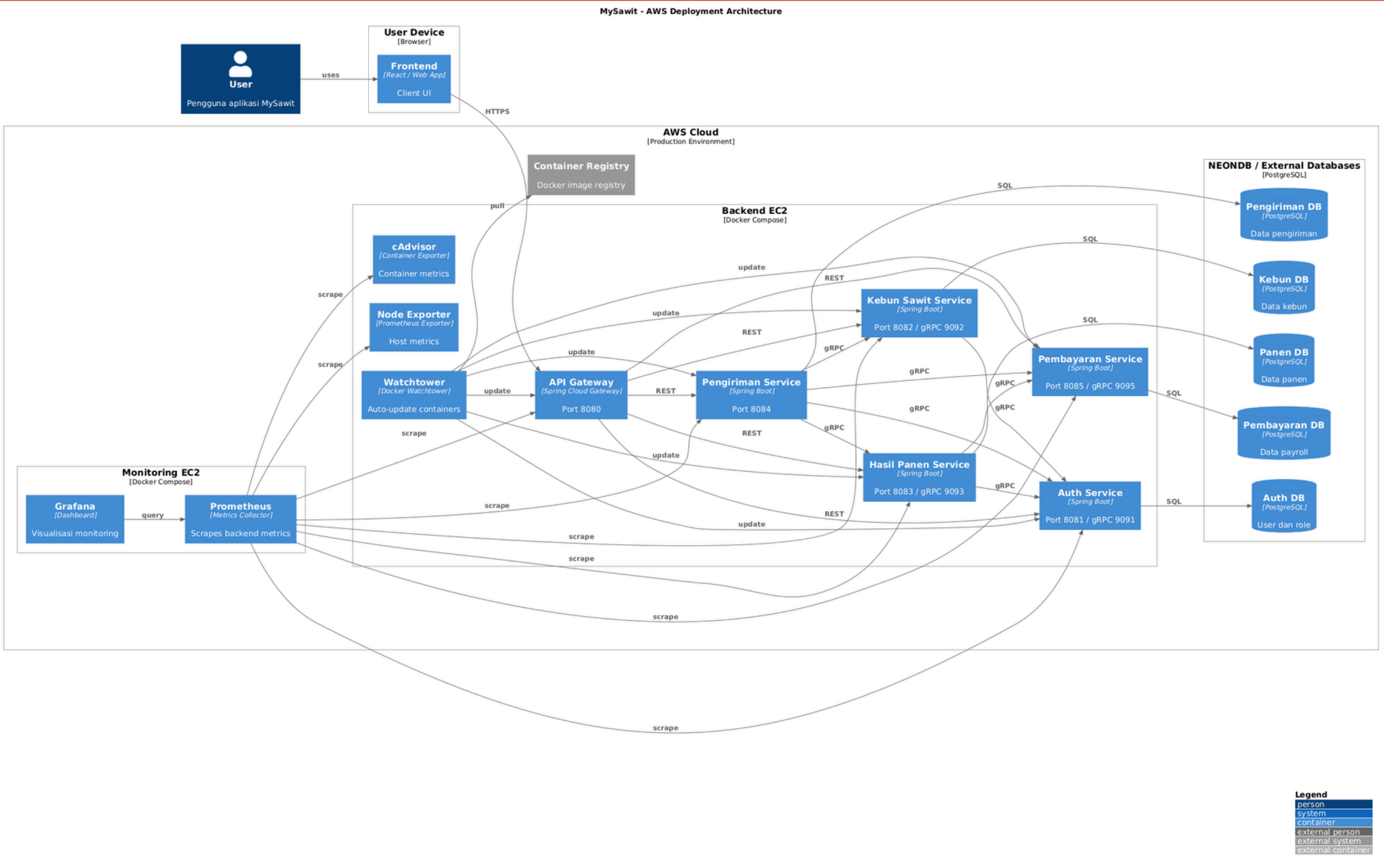
sawit-panen

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
 id.ac.ui.cs.advprog.sawitpanen.service		90%		66%	17	60	27	257	14	51	0	3
 id.ac.ui.cs.advprog.sawitpanen.repository		98%		100%	1	19	0	79	1	14	0	3
 id.ac.ui.cs.advprog.sawitpanen.internal.grpc		100%		100%	0	93	0	407	0	76	0	9
 id.ac.ui.cs.advprog.sawitpanen.controller		100%		n/a	0	12	0	76	0	12	0	2
 id.ac.ui.cs.advprog.sawitpanen.exception		100%		n/a	0	13	0	41	0	13	0	4
 id.ac.ui.cs.advprog.sawitpanen.dto		100%		n/a	0	2	0	12	0	2	0	1
 id.ac.ui.cs.advprog.sawitpanen.model		100%		n/a	0	2	0	5	0	2	0	2
 id.ac.ui.cs.advprog.sawitpanen		100%		n/a	0	6	0	8	0	6	0	2
Total	129 of 4,025	96%	6 of 62	90%	18	207	27	885	15	176	0	26

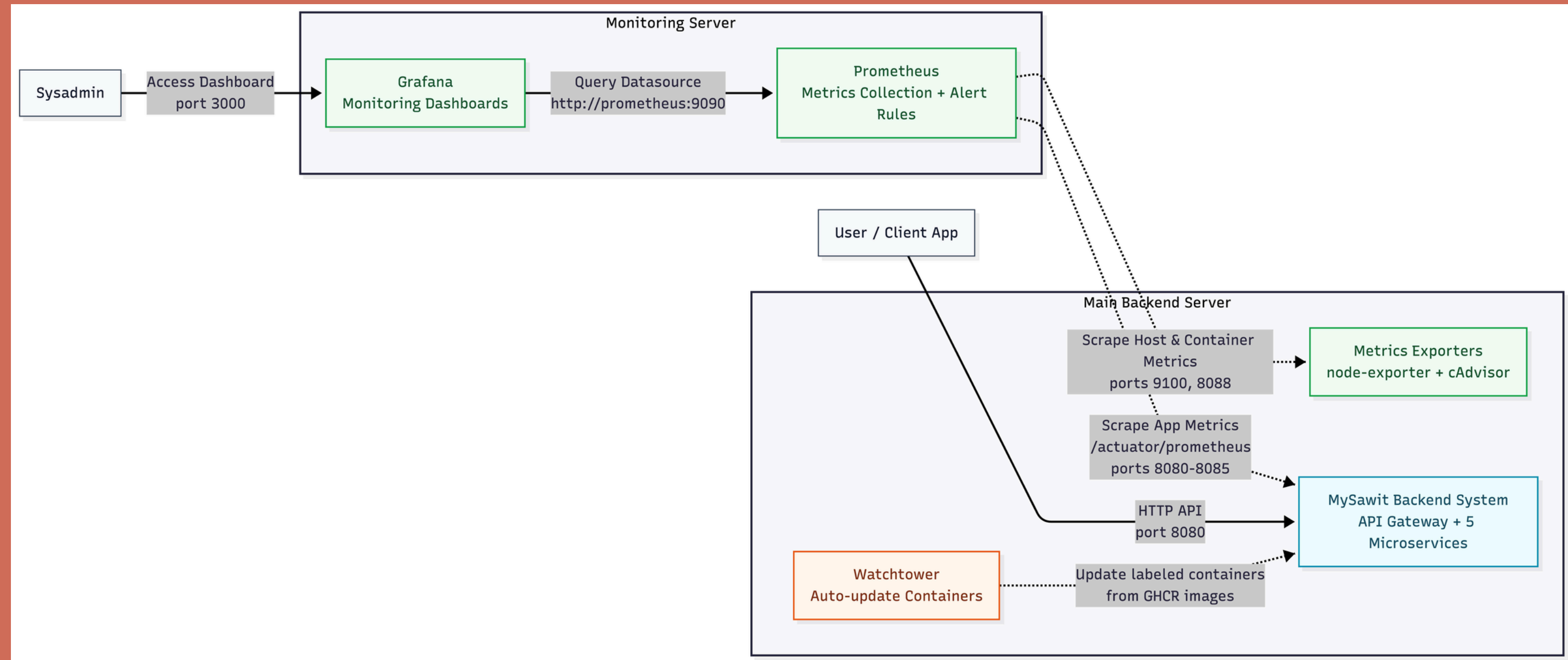
JaCoCo Report

Software Architecture

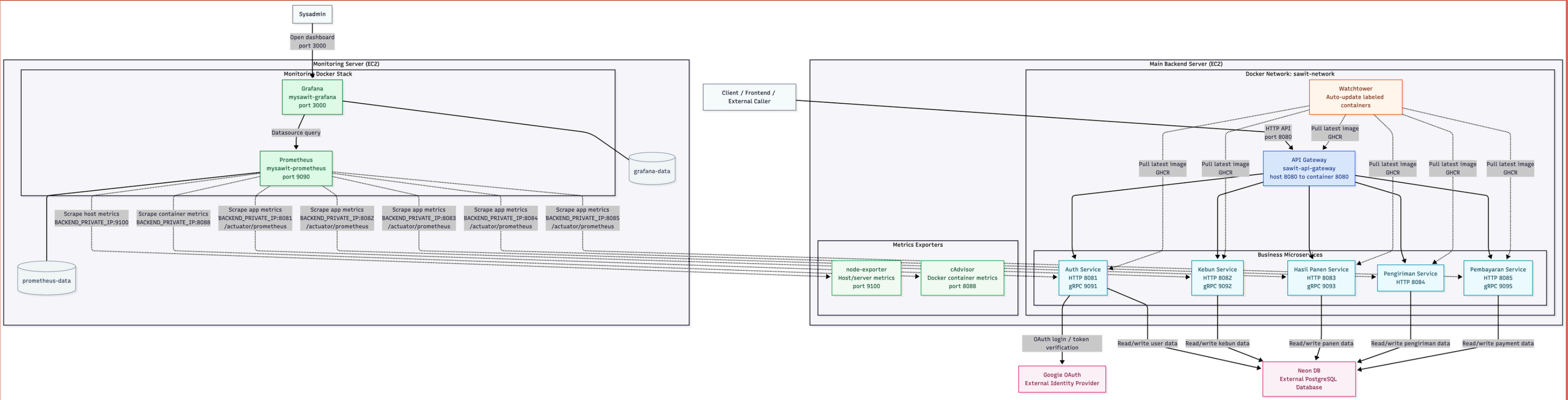
Microservice



Software Architecture Microservice

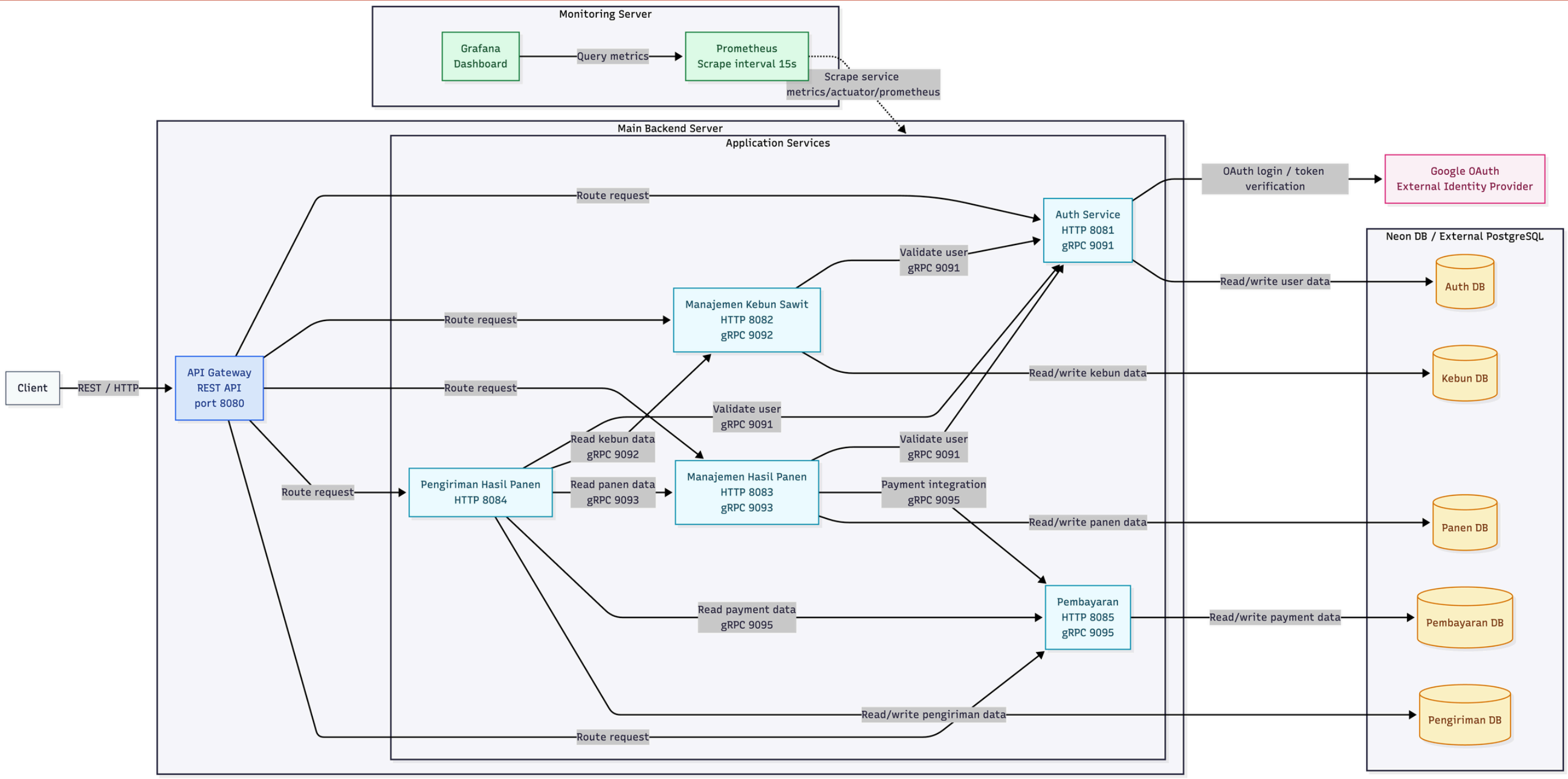


Software Architecture Microservice



Software Architecture

Microservice



Software Architecture

REST API



00 Auth

- POST** Register
- POST** Login Local
- POST** Login Google
- POST** Refresh Token
- POST** Logout
- GET** Get My Profile
- PUT** Update My Profile
- GET** Admin Search Users
- GET** Admin Get User Detail
- PUT** Admin Assign Buruh To Mandor
- PUT** Admin Unassign Buruh Mandor
- DEL** Admin Delete User
- GET** Mandor Get Bawahan

01 Kebun Sawit

- GET** List Kebun
- GET** Get Kebun By Kode
- POST** Create Kebun
- PUT** Update Kebun
- DEL** Delete Kebun
- POST** Assign Mandor To Kebun
- DEL** Unassign Mandor From Kebun
- POST** Assign Supir To Kebun
- DEL** Unassign Supir From Kebun

02 Hasil Panen

- POST** Create Laporan Panen
- GET** List Panen
- GET** Get Panen Detail
- PATCH** Approve Panen
- PATCH** Reject Panen

Software Architecture

REST API



✓ 03 Pengiriman Hasil Panen

- POST Create Pengiriman / Tugaskan...
- GET List Pengiriman
- PUT Update Status Pengiriman
- GET Supir List Pengiriman Saya
- GET Mandor List Pengiriman Kebun
- GET Mandor List Pengiriman Supir ...
- GET Mandor List Supir Satu Kebun
- PUT Mandor Approve Pengiriman
- PUT Mandor Reject Pengiriman
- GET Admin List Pengiriman Disetuj...
- PUT Admin Approve Pengiriman
- PUT Admin Reject Pengiriman
- PUT Admin Partial Reject Pengiriman

✓ 04 Pembayaran

- PUT Set Payroll Rates
- GET Get Current Payroll Rates
- GET My Payrolls By Owner
- GET Admin List All Payrolls
- GET Get Wallet By Owner
- GET Get Transaction History
- POST Accept Payroll
- POST Reject Payroll
- POST Pay Payroll
- POST Create Admin Wallet Top Up
- GET List Top Up Requests
- POST Confirm Top Up Request
- POST Internal Event - Harvest Appro...
- POST Internal Event - Shipment Man...
- POST Internal Event - Shipment Ad...

Software Architecture

GRPC



Caller	Target	RPC	Tujuan
Kebun	Auth	GetUserById	Ambil nama mandor/supir valid
Kebun	Auth	ValidateUserRole	Pastikan role mandor/supir benar
Panen	Auth	ValidateUserRole	Pastikan mandor/buruh valid
Panen	Pembayaran	ProcessHarvestApproved	Payroll buruh setelah panen approved
Pengiriman	Auth	GetUserById	Tampilkan nama mandor/supir
Pengiriman	Auth	ValidateUserRole	Validasi mandor/supir
Pengiriman	Kebun	GetKebunByMandorId	Ambil kebun mandor
Pengiriman	Kebun	GetKebunBySupirId	Ambil kebun supir
Pengiriman	Kebun	ValidateMandorSupirSameKebun	Validasi assignment
Pengiriman	Kebun	GetSupirByKebun	List supir satu kebun
Pengiriman	Panen	GetPanenByIds	Ambil detail hasil panen
Pengiriman	Panen	ValidatePanenApproved	Pastikan panen sudah approved
Pengiriman	Pembayaran	ProcessShipmentMandorApproved	Payroll supir
Pengiriman	Pembayaran	ProcessShipmentAdminApproved	Payroll mandor
Pembayaran	Auth	GetUserById opsional	Tampilkan owner payroll

Software Development CI/CD (SERVICES/GATEWAY)



```
1 name: CI/CD Pipeline API Gateway
2 on:
3   push:
4     branches: ["main"]
5   pull_request:
6     branches: ["main"]
7
8 jobs:
9   Continuous-Integration:
10    runs-on: ubuntu-22.04
11    steps:
12      - name: Check out the Git repository
13        uses: actions/checkout@v4
14        with:
15          fetch-depth: 0
16
17      - name: Set up Java toolchain
18        uses: actions/setup-java@v4
19        with:
20          distribution: "temurin"
21          java-version: "21"
22          cache: "gradle"
23
24      - name: Grant execute permission for gradlew
25        run: chmod +x gradlew
26
27      - name: Run unit tests
28        run: ./gradlew test
```

```
Continuous-Deployment:
  needs: Continuous-Integration
  runs-on: ubuntu-22.04
  if: github.ref == 'refs/heads/main' && github.event_name == 'push'
  permissions:
    contents: read
    packages: write
  steps:
    - name: Check out the Git repository
      uses: actions/checkout@v4
    - name: Set up Java toolchain
      uses: actions/setup-java@v4
      with:
        distribution: "temurin"
        java-version: "21"
        cache: "gradle"
    - name: Build JAR File
      run: |
        chmod +x gradlew
        ./gradlew clean build -x test
```

```
- name: Log in to GHCR
  uses: docker/login-action@v3
  with:
    registry: ghcr.io
    username: ${github.actor}
    password: ${secrets.GITHUB_TOKEN}
- name: Build and push Docker image
  uses: docker/build-push-action@v5
  with:
    context: .
    push: true
    tags: ghcr.io/advprog-2026-a18-project/api-sawit-gateway:latest
```

Software Development CI/CD (DEPLOYMENT)



```
name: Continuous Deployment (AWS)

on:
  push:
    branches: [ "main" ]

jobs:
  deploy:
    runs-on: ubuntu-latest
    permissions:
      packages: read
      contents: read

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Deploy to AWS via SSH
        uses: appleboy/ssh-action@master
        with:
          host: ${ secrets.REMOTE_HOST }
          username: ${ secrets.REMOTE_USER }
          key: ${ secrets.SSH_PRIVATE_KEY }
          envs: GITHUB_TOKEN
          script: |
            cd ~/mysawit-deployment || exit

            # update file docker-compose.yml
            git pull origin main

            # login to ghcr
            echo "${ secrets.GITHUB_TOKEN }" | sudo docker login ghcr.io -u ${ github.actor } --password-stdin

            # pull image baru
            sudo docker-compose pull

            # restart containers
            sudo docker-compose up -d --remove-orphans

            # cleaning
            sudo docker image prune -f
```

Thank You

Any Question?

```
import id.ac.ui.cs.advprog.sawitpanen.model.Panen;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.data.jpa.repository.JpaSpecificationExecutor;
import org.springframework.stereotype.Repository;

import java.time.LocalDate;
import java.util.List;
import java.util.UUID;

@Repository 14 usages ⓘ FrankyRay +1
public interface PanenRepository extends JpaRepository<Panen, UUID>, JpaSpecificationExecutor<Panen> {
    boolean existsByBuruhIdAndTanggalPanen(UUID buruhId, LocalDate tanggalPanen); 5 usages ⓘ FrankyRay
    List<Panen> findByBuruhId(UUID buruhId); 2 usages ⓘ evanhwz
}
```